

A.2. Main results

We will require the following assumptions. These are fairly agnostic as to the kernel form; Appendix A.4.2 shows that these assumptions hold (and gives the constants) for the kernels (1) we use in the paper.

(A) The kernels k_ω are uniformly bounded:

$$\sup_{\omega \in \Omega} \sup_{x \in \mathcal{X}} k_\omega(x, x) \leq \nu.$$

For the kernels we use in practice, $\nu = 1$.

(B) The possible kernel parameters ω lie in a Banach space of dimension D . Furthermore, the set of possible kernel parameters Ω is bounded by R_ω , $\Omega \subseteq \{\omega \mid \|\omega\| \leq R_\Omega\}$.

Appendix A.4.2 builds this space and its norm for the kernels we use in the paper.

(C) The kernel parameterization is Lipschitz: for all $x, y \in \mathcal{X}$ and $\omega, \omega' \in \Omega$,

$$|k_\omega(x, y) - k_{\omega'}(x, y)| \leq L_k \|\omega - \omega'\|.$$

Proposition 23 in Appendix A.4.2 gives an expression for L_k for the kernels we use in the paper.

We will first show the main results under these general assumptions, using uniform convergence results shown in Appendix A.3, then show Assumptions (B) and (C) for particular kernels in Appendix A.4.2.

Theorem 11. *Under Assumptions (A) to (C), let $\bar{\Omega}_s \subseteq \Omega$ be the set of kernel parameters for which $\sigma_\omega^2 \geq s^2$, and assume $\nu \geq 1$. Take $\lambda = n^{-1/3}$. Then, with probability at least $1 - \delta$,*

$$\sup_{\omega \in \bar{\Omega}_s} \left| \frac{\hat{\eta}_\omega}{\hat{\sigma}_{\omega, \lambda}} - \frac{\eta_\omega}{\sigma_\omega} \right| \leq \frac{2\nu}{s^2 n^{1/3}} \left(\frac{1}{s} + \frac{2304\nu^2}{\sqrt{n}} + \left\lceil \frac{4s}{n^{1/6}} + 1024\nu \right\rceil \left[L_k + \sqrt{2 \log \frac{2}{\delta} + 2D \log(4R_\Omega \sqrt{n})} \right] \right),$$

and thus, treating ν as a constant,

$$\sup_{\omega \in \bar{\Omega}_s} \left| \frac{\hat{\eta}_\omega}{\hat{\sigma}_{\omega, \lambda}} - \frac{\eta_\omega}{\sigma_\omega} \right| = \tilde{O}_P \left(\frac{1}{s^2 n^{1/3}} \left[\frac{1}{s} + L_k + \sqrt{D} \right] \right).$$

Proof. Let $\sigma_{\omega, \lambda}^2 := \sigma_\omega^2 + \lambda$. Using $|\hat{\eta}_\omega| \leq 4\nu$, we begin by decomposing

$$\begin{aligned} \sup_{\omega \in \bar{\Omega}_s} \left| \frac{\hat{\eta}_\omega}{\hat{\sigma}_{\omega, \lambda}} - \frac{\eta_\omega}{\sigma_\omega} \right| &\leq \sup_{\omega \in \bar{\Omega}_s} \left| \frac{\hat{\eta}_\omega}{\hat{\sigma}_{\omega, \lambda}} - \frac{\hat{\eta}_\omega}{\sigma_{\omega, \lambda}} \right| + \sup_{\omega \in \bar{\Omega}_s} \left| \frac{\hat{\eta}_\omega}{\sigma_{\omega, \lambda}} - \frac{\hat{\eta}_\omega}{\sigma_\omega} \right| + \sup_{\omega \in \bar{\Omega}_s} \left| \frac{\hat{\eta}_\omega}{\sigma_\omega} - \frac{\eta_\omega}{\sigma_\omega} \right| \\ &= \sup_{\omega \in \bar{\Omega}_s} |\hat{\eta}_\omega| \frac{1}{\hat{\sigma}_{\omega, \lambda}} \frac{1}{\sigma_{\omega, \lambda}} \frac{|\hat{\sigma}_{\omega, \lambda}^2 - \sigma_{\omega, \lambda}^2|}{\hat{\sigma}_{\omega, \lambda} + \sigma_{\omega, \lambda}} + \sup_{\omega \in \bar{\Omega}_s} |\hat{\eta}_\omega| \frac{1}{\sigma_{\omega, \lambda}} \frac{1}{\sigma_\omega} \frac{|\sigma_{\omega, \lambda}^2 - \sigma_\omega^2|}{\sigma_{\omega, \lambda} + \sigma_\omega} + \sup_{\omega \in \bar{\Omega}_s} \frac{1}{\sigma_\omega} |\hat{\eta}_\omega - \eta_\omega| \\ &\leq \sup_{\omega \in \bar{\Omega}_s} \frac{4\nu}{\sqrt{\lambda} s (s + \sqrt{\lambda})} |\hat{\sigma}_\omega^2 - \sigma_\omega^2| + \frac{4\nu \lambda}{\sqrt{s^2 + \lambda} s (\sqrt{s^2 + \lambda} + s)} + \sup_{\omega \in \bar{\Omega}_s} \frac{1}{s} |\hat{\eta}_\omega - \eta_\omega| \\ &\leq \frac{4\nu}{s^2 \sqrt{\lambda}} \sup_{\omega \in \bar{\Omega}} |\hat{\sigma}_\omega^2 - \sigma_\omega^2| + \frac{2\nu}{s^3} \lambda + \frac{1}{s} \sup_{\omega \in \bar{\Omega}} |\hat{\eta}_\omega - \eta_\omega|. \end{aligned}$$

Propositions 15 and 16 show uniform convergence of $\hat{\eta}_\omega$ and $\hat{\sigma}_\omega^2$, respectively. Thus, with probability at least $1 - \delta$, the error is at most

$$\frac{2\nu}{s^3} \lambda + \left[\frac{8\nu}{s\sqrt{n}} + \frac{1792\nu}{\sqrt{n}s^2\sqrt{\lambda}} \right] \sqrt{2 \log \frac{2}{\delta} + 2D \log(4R_\Omega \sqrt{n})} + \left[\frac{8}{s\sqrt{n}} + \frac{2048\nu^2}{\sqrt{n}s^2\sqrt{\lambda}} \right] L_k + \frac{4608\nu^3}{s^2 n \sqrt{\lambda}}.$$

Taking $\lambda = n^{-1/3}$ gives

$$\frac{2\nu}{s^3 n^{1/3}} + \left[\frac{8\nu}{s\sqrt{n}} + \frac{1792\nu}{s^2 n^{1/3}} \right] \sqrt{2 \log \frac{2}{\delta} + 2D \log(4R_\Omega \sqrt{n})} + \left[\frac{8}{s\sqrt{n}} + \frac{2048\nu^2}{s^2 n^{1/3}} \right] L_k + \frac{4608\nu^3}{s^2 n^{5/6}}.$$

Using $1 \leq \nu$, $1792 < 2048$, we can get the slightly simpler upper bound

$$\frac{2\nu}{s^3 n^{1/3}} + \left[\frac{8\nu}{s\sqrt{n}} + \frac{2048\nu^2}{s^2 n^{1/3}} \right] \left[L_k + \sqrt{2 \log \frac{2}{\delta} + 2D \log(4R_\Omega \sqrt{n})} \right] + \frac{4608\nu^3}{s^2 n^{5/6}}. \quad \square$$

It is worth noting that, if we are particularly concerned about the s dependence, we can make some slightly different choices in the decomposition to improve the dependence on s while worsening the rate with n .

Corollary 12. *In the setup of Theorem 11, additionally assume that there is a unique population maximizer ω^* of J from (3), i.e. for each $t > 0$ we have*

$$\sup_{\omega \in \bar{\Omega}_s : \|\omega - \omega^*\| \geq t} J(\mathbb{P}, \mathbb{Q}; k_\omega) < J(\mathbb{P}, \mathbb{Q}; k_{\omega^*}).$$

For each n , let $S_{\mathbb{P}}^{(n)}$ and $S_{\mathbb{Q}}^{(n)}$ be sequences of sample sets of size n , let $\hat{J}_n(\omega)$ denote $J_{\lambda=n^{-1/3}}(S_{\mathbb{P}}^{(n)}, S_{\mathbb{Q}}^{(n)}; k_\omega)$, and take $\hat{\omega}_n^*$ to be a maximizer of $\hat{J}_n(\omega)$.⁷ Then $\hat{\omega}_n^*$ converges in probability to ω^* .

Proof. By Theorem 11, $\sup_{\omega \in \bar{\Omega}_s} |\hat{J}_n(\omega) - J(\omega)| \xrightarrow{P} 0$. Then the result follows by Theorem 5.7 of Van der Vaart (2000). $\square$

Corollary 13. *In the setup of Theorem 11, suppose we use n sample points to select a kernel $\hat{\omega}_n \in \arg \max_{\omega \in \bar{\Omega}_s} \hat{J}_\lambda(\omega)$ and m sample points to run a test of level α . Let $r_{\hat{\omega}_n}^{(m)}$ denote the rejection threshold for a test with that kernel of size m . Define $J^* := \sup_{\omega \in \bar{\Omega}_s} J(\omega)$, and constants C, C', C'', N_0 depending on ν, L_k, D, R_Ω and s . For any $n \geq N_0$, with probability at least $1 - \delta$, this test procedure has power*

$$\Pr \left(m \hat{\omega}_n > r_{\hat{\omega}_n}^{(m)} \right) \geq \Phi \left(\sqrt{m} J^* - C \frac{\sqrt{m}}{n^{1/3}} \sqrt{\log \frac{n}{\delta}} - C' \sqrt{\log \frac{1}{\alpha}} \right) - \frac{C''}{\sqrt{m}}.$$

Proof. Let $\hat{\omega}_n \in \arg \max_{\omega \in \bar{\Omega}_s} \hat{J}_\lambda(\omega)$. By Theorem 11, there are some N_0, C depending on ν, L_k, D, R_Ω , and s such that as long as $n \geq N_0$, with probability at least $1 - \delta$ it holds that

$$\sup_{\omega \in \bar{\Omega}_s} |J_\lambda(\omega) - J(\omega)| \leq \frac{1}{2} C n^{-1/3} \sqrt{\log \frac{n}{\delta}} =: \epsilon_n.$$

Assume for the remainder of this proof that this event holds. Letting $\omega^* \in \arg \max J(\omega)$, we know because $\hat{\omega}_n$ maximizes $\hat{J}_\lambda$ that $\hat{J}_\lambda(\hat{\omega}_n) \geq \hat{J}_\lambda(\omega^*)$. Using uniform convergence twice,

$$J(\hat{\omega}_n) \geq \hat{J}_\lambda(\hat{\omega}_n) - \epsilon_n \geq \hat{J}_\lambda(\omega^*) - \epsilon_n \geq (J(\omega^*) - \epsilon_n) - \epsilon_n = J^* - 2\epsilon_n.$$

Now, although Proposition 2 establishes that $r_\omega^{(m)} \rightarrow r_\omega$ and it is even known (Korolyuk & Borovskikh, 1988, Theorem 5) that $|r_\omega^{(m)} - r_\omega|$ is $o(1/\sqrt{m})$, the constant in that convergence will depend on the choice of ω in an unknown way. It's thus simpler to use the very loose but uniform (McDiarmid-based) bound given by Corollary 11 of Gretton et al. (2012a), which implies $r_\omega^{(m)} \leq 4\nu \sqrt{\log(\alpha^{-1})m}$ no matter the choice of ω .

We will now need a more precise characterization of the power than that provided by the central limit theorem of Proposition 2. Callaert & Janssen (1978) provide such a result, a Berry-Esseen bound on U -statistic convergence: there is some absolute constant $C'_{BS} = 2^3 4^3 C_{BS}$ such that

$$\sup_t \left| \Pr \left(\sqrt{m} \frac{\hat{\eta}_\omega - \eta_\omega}{\sigma_\omega^2} \leq t \right) - \Phi(t) \right| \leq \frac{C'_{BS} \mathbb{E}|H_{12}|^3}{(\sigma_\omega/2)^3 \sqrt{m}} \leq \frac{C_{BS} \nu^3}{\sigma_\omega^3 \sqrt{m}}.$$

⁷In fact, it suffices for the $\hat{\omega}_n^*$ to only approximately maximize $\hat{J}_n$, as long as their suboptimality is $o_P(1)$.

Letting $r_\omega^{(m)}$ be the appropriate rejection threshold for k_ω with m samples, the power of a test with kernel k_ω is

$$\begin{aligned} \Pr \left(m\hat{\eta}_\omega > r_\omega^{(m)} \right) &= \Pr \left(\sqrt{m} \frac{\hat{\eta}_\omega - \eta_\omega}{\sigma_\omega} > \frac{r_\omega^{(m)}}{\sqrt{m}\sigma_\omega} - \sqrt{m} \frac{\eta_\omega}{\sigma_\omega} \right) \\ &\geq \Phi \left(\sqrt{m} J(\omega) - \frac{r_\omega^{(m)}}{\sqrt{m}\sigma_\omega} \right) - \frac{C_{BS}\nu^3}{\sigma_\omega^3 \sqrt{m}} \\ &\geq \Phi \left(\sqrt{m} J(\omega) - \frac{r_\omega^{(m)}}{s\sqrt{m}} \right) - \frac{C''}{\sqrt{m}}, \end{aligned}$$

using a new constant $C'' := C_{BS}\nu^3/s^3$. Combining the previous results on $J(\hat{\omega}_n)$ and $r_{\hat{\omega}_n}^{(m)}$ yields the claim. $\square$

Corollary 14. *In the setup of Corollary 13, suppose we are given N data points to divide between n training points and $m = N - n$ testing points, and $\delta < 0.22$ is fixed. Ignoring the Berry-Esseen convergence term outside of Φ , the asymptotic power upper bound*

$$\Phi \left(\sqrt{m} J^* - C \frac{\sqrt{m}}{n^{\frac{1}{3}}} \sqrt{\log \frac{n}{\delta}} - C' \sqrt{\log \frac{1}{\alpha}} \right)$$

is maximized only when, as other quantities remain constant, we pick n to satisfy

$$\lim_{N \rightarrow \infty} \frac{n}{\left(\frac{C}{\sqrt{3}J^*} N \sqrt{\log N} \right)^{\frac{3}{4}}} = 1.$$

Proof. Because the C' term is constant, we wish to choose

$$\arg \max_{0 < n < N} \frac{J^*}{C} \sqrt{N-n} - \frac{\sqrt{N-n}}{n^{\frac{1}{3}}} \sqrt{\log \frac{n}{\delta}}.$$

Clearly neither endpoint is optimal. Relaxing n to be real-valued, the optimum must be achieved at a stationary point, where

$$\frac{-J^*}{2C\sqrt{N-n}} + \frac{\sqrt{\log \frac{n}{\delta}}}{2\sqrt{N-n}n^{\frac{1}{3}}} + \frac{1}{3}\sqrt{N-n}n^{-\frac{4}{3}}\sqrt{\log \frac{n}{\delta}} - \frac{1}{2}\sqrt{N-n}n^{-\frac{4}{3}}\left(\log \frac{n}{\delta}\right)^{-\frac{1}{2}} = 0.$$

Multiplying by $2\sqrt{N-n}n^{\frac{4}{3}}\sqrt{\log \frac{n}{\delta}}$ and rearranging, we get that a stationary point is achieved exactly when

$$\underbrace{\frac{1}{3}[n+2N]\log \frac{n}{\delta} + n}_D = \underbrace{\frac{J^*}{C}n^{\frac{4}{3}}\sqrt{\log \frac{n}{\delta}} + N}_E.$$

Now write, without loss of generality, $n = (A_N N \sqrt{\log N})^{\frac{3}{4}}$, and so

$$\begin{aligned} D &= \frac{1}{3} \left[A_N^{\frac{3}{4}} N^{\frac{3}{4}} (\log N)^{\frac{3}{8}} + 2N \right] \left[\underbrace{\frac{3}{4} \log A_N + \frac{3}{4} \log N + \frac{3}{8} \log \log N + \log \frac{1}{\delta}}_{\log n} \right] + A_N^{\frac{3}{4}} N^{\frac{3}{4}} (\log N)^{\frac{3}{8}} \\ E &= \frac{J^*}{C} A_N N \sqrt{\log N} \sqrt{\underbrace{\frac{3}{4} \log A_N + \frac{3}{4} \log N + \frac{3}{8} \log \log N + \log \frac{1}{\delta}}_{\log n}} + N. \end{aligned}$$

We will show that $D - E \rightarrow 0$ requires $A_N \rightarrow C/(\sqrt{3}J^*)$, implying the result.